Lesson-1 Q&A — rates vs. morphing, profiling, and the anatomy of learned ratios

teaching companion note — Q&A of 11 July 2026, organized

11 July 2026

What this note is

A cleaned-up, *complete* record of the Q&A session revisiting Lesson 1 (the NSBI-systematics field map), grouped logically rather than chronologically. Part 1: what in the likelihood is *known a priori* vs. *learned*, and what the learned objects actually are (the green rate factor; weights vs. density ratios; κ_λ-independence; the per-event anchor construction). Part 2: what profiling is and exactly how a tilted κ_λ–α likelihood widens the confidence interval, plus the meaning of the profile likelihood ratio $\lambda(\kappa_\lambda)$. Part 3: the four off-shell (R2) modeling assumptions unpacked, and how each breaks in $b\bar{b}\tau\tau$, $b\bar{b}\gamma\gamma$, and $b\bar{b}b\bar{b}$. Part 4: how R3 (Gaussian-Process morphing) and R4 (the "refinable" surrogate) relax those assumptions — including the verified correction that R4 is *not* a generative/normalizing-flow method. Companion to the κ_λ-challenges note (`2026-06-20-klambda-nsbi-vs-offshell-challenges`); challenge labels C1–C6 refer to its Sec. 5.

Contents

1 The model's anatomy: what is known, what is learned

1.1 "The green factor κ (rates) has a ***known a-priori*** dependence"

Question

Can you elaborate "The green factor κ (rates: lumi, cross-section, POIs) has a known a-priori dependence"?

In the Lesson-1 master equation, each sample s contributes a product of four factors, and the color-coding splits them by *where the parameter-dependence comes from*:

$$P(x \mid \eta, \chi) \;\propto\; \prod_{\text{events } e} \Bigg[\sum_{\text{samples } s} \nu_s(\eta, \xi) \cdot \kappa_s(\eta, \xi) \cdot \boldsymbol{\Delta_s(x_e \mid \alpha)} \cdot P_s(x_e) \Bigg] \cdot \prod_p f(a_p|\chi_p), \tag{1}$$

with η the POIs, ξ unconstrained normalization parameters, α the constrained nuisance parameters, and the last factor the Gaussian constraint terms.

- $P_s(x)$ — the *nominal* per-event shape of sample s. Fixed; learned once (the ratio network at $\alpha = 0$).
- $\nu_s \cdot \kappa_s$ (**green**) — the *expected event count* of sample s and its multiplicative rate modifiers: yield = luminosity $\times$ cross-section $\times$ efficiency, scaled by the POI and by normalization nuisances.
- $\Delta_s(x|\alpha)$ (**red**) — how the *shape* deforms when a systematic is dialed.

"Known a-priori" means: for every parameter entering the green factor, the functional form is something you can **write down in closed form before touching any simulation**, with exact derivatives — there is nothing to learn:

- **Luminosity.** The yield scales *linearly* in L ($\nu \propto L$); a $\pm 1.6\%$ lumi NP multiplies every MC sample's rate by the log-normal $e^{\alpha\sigma_{\ln}}$. Pure bookkeeping.
- **Cross-section / normalization NPs.** The same closed log-normal form, per process.
- **POIs.** For the off-shell measurement, the mixture coefficients $(\mu - \sqrt{\mu},\ \sqrt{\mu},\ 1 - \sqrt{\mu})$ are exact functions of μ from the amplitude structure. For κ_λ:

  $$\sigma(\kappa_\lambda) \;=\; \kappa_\lambda^2 A \;+\; \kappa_\lambda B \;+\; C, \tag{2}$$

  an *exact parabola* from $|\kappa_\lambda \cdot \text{triangle} + \text{box}|^2$; simulation only supplies the three constants A, B, C.

The deeper point for κ_λ: **even the POI's *shape* dependence is handled a-priori** via the mixture trick. One never learns "how the shape morphs with κ_λ" — the signal is split into three fixed-shape components ($|\text{triangle}|^2$, interference, $|\text{box}|^2$) with fixed $P_s(x)$, and *all* κ_λ-dependence lives in the known green coefficients ($\kappa_\lambda^2,\ \kappa_\lambda,\ 1$). The learning budget is reserved for the red factor — the systematics — where no amplitude-level decomposition exists: the α-dependence of the shape emerges from the shower$\to$detector$\to$reconstruction chain, has no closed form, and is intractable. That asymmetry — **POI dependence by construction, nuisance dependence by learning** — is the core architecture of the whole method.

Caveat: the green/red boundary is where C4 lives

The green/red split is a *modeling choice*, and it can be wrong. `THU_HH` is naively filed under green (a ±lnN rate modifier), but faithfully it is κ_λ-dependent *and* shape-bearing (the three signal components carry different theory bands) — so it leaks across the boundary into red and, worse, couples POI×nuisance. "Which factor does this NP really belong in?" is a useful discipline for every systematic in the table.

1.2 Weights *are* density ratios: the two learned families

Question

We've been talking about learning density ratios — what do you mean by learning a *weight per component*?

The two phrases sound like different machinery, and they are not. **A "weight" *is* a learned density ratio.** The only thing that changes is *which two densities* sit in the ratio. The full NSBI model contains **two families of learned ratios**, distinguished purely by their numerator/denominator (Fig. 1):

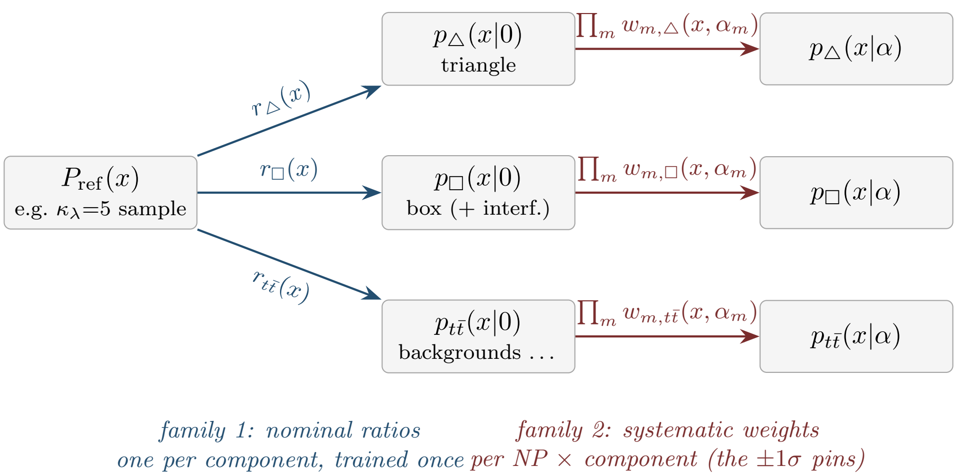

Figure 1: The two families of learned density ratios. Every arrow is the *same kind of object* — a density ratio learned by a classifier — differing only in which two samples it compares. The mixture then combines components with the known coefficients: $\nu(x;\kappa_\lambda,\alpha) = \sum_s c_s(\kappa_\lambda)\,\nu_s\,r_s(x)\prod_m w_{m,s}(x,\alpha_m)\,P_{\mathrm{ref}}(x)$, with $c_s = (\kappa_\lambda^2, \kappa_\lambda, 1, \ldots)$.

"Learning a weight per component" therefore means: for one NP, say JES, one trains a separate Family-2 ratio for each middle-column box of Fig. 1 — one classifier for (JES+1σ triangle vs. nominal triangle), another for the box component, another for $t\bar{t}$, and so on. Each is a density ratio in exactly the sense used throughout. The pins of Sec. 1.4 are the *values of one Family-2 ratio at* $\alpha = \pm 1$ for one event.

Why it is called a "weight" rather than a "ratio". Because of how it is *used*: multiplying the nominal density by it *is* per-event reweighting — event e, carrying weight $w(x_e,\alpha)$, emulates the varied sample without regenerating it (the CARL/DCTR reweighting sense of Lesson 7). Math-

	Family 1: nominal ratio $r_s(x)$	**Family 2: systematic weight $w_{m,s}(x, \alpha_m)$**
the ratio	$p_s(x\|\alpha{=}0)$ / $P_{\mathrm{ref}}(x)$	$p_s(x\|\alpha_m{=}\pm1)$ / $p_s(x\|\alpha{=}0)$
compares	*different components*, both nominal	*the same component*, varied vs nominal
trained by	classifier: component-s sample **vs reference** $\to s/(1{-}s)$	classifier: $\pm1\sigma$-varied sample of s **vs nominal** of $s \to s/(1{-}s)$
carries	the physics separation (κ_λ info via the mixture)	the systematic deformation (the α-dependence)
how many	one per component ($\sim$5–7)	$2 \times N_{\mathrm{NP}} \times N_{\mathrm{comp}}$ ($\sim$thousands) — this is C2
typical size	can be large/structured (different processes)	$\approx$ 1 everywhere (varied $\approx$ nominal): small multiplicative corrections

Table 1: Same trick, different sample pairs.

ematically a ratio; operationally a weight. The chain composes by multiplication:

$$p_s(x|\alpha) \;=\; w_s(x,\alpha) \cdot p_s(x|0) \;=\; w_s(x,\alpha) \cdot r_s(x) \cdot P_{\mathrm{ref}}(x) \tag{3}$$

— two hops: reference → component nominal (blue, Family 1), then nominal → varied (red, Family 2).

Why factor into two hops instead of learning $p_s(x|\alpha)/P_{\mathrm{ref}}$ directly as one α-parameterized ratio? Three practical reasons:

1. **Statistics where it matters.** The Family-1 ratio is hard (genuinely different distributions) but is trained *once*, on the full nominal MC (the ample ≈500–750k events per κ_λ point). The Family-2 ratios are *easy* (two nearly identical distributions, ratio ≈ 1) — which is what makes them learnable at all from the smaller $\pm1\sigma$ samples.

2. **Modularity.** Adding, dropping, or retraining an NP never touches the nominal ratios or any other NP's weights.

3. **It is the factorization ansatz made flesh** — one Family-2 curve per NP, multiplied — with all the costs discussed in Sec. 3.

Anchor in the project's own code. The preliminary analysis (`DiHiggs_Lambda_NSBI`) has built **only Family 1** — the ensembles of per-basis-sample and per-background ratios against the $\kappa_\lambda{=}5$ reference. It has zero Family-2 objects, which is precisely the operational meaning of "no systematic NPs in the fit yet." The systematics program (C1/C2/C5/C6) is, concretely, the project of standing up the red-arrow family — thousands of near-unity ratios — and controlling their interpolation, cross-terms, and MC-statistical noise.

1.3 Are the weights κ_λ-independent? (yes — with one sharpening)

Question

Most systematics weights are λ-independent (except the theory systematics), right?

Yes — and by *design*, not by accident. In the factorized scheme every systematic weight takes only (x, α_m) as arguments; κ_λ never appears inside any of them. The κ_λ-dependence of the *total* prediction lives entirely upstairs, in the known mixture coefficients:

$$\nu(x;\kappa_\lambda,\alpha) \;=\; \sum_{\text{components } s} c_s(\kappa_\lambda)\cdot\nu_s\cdot\big[\text{nominal ratio}_s(x)\big]\cdot\prod_m w_{m,s}(x,\alpha_m), \qquad c_s=(\kappa_\lambda^2,\ \kappa_\lambda,\ 1). \quad (4)$$

For **experimental** NPs this κ_λ-independence is *physically correct* — the detector does not know the coupling; a $+1\sigma$ JES miscalibration does the same thing to a given event whatever κ_λ is. But note the load-bearing subtlety: **this only holds because the weights are learned per component.** JES deforms the triangle, box, and interference densities *differently* (they have different m_{HH} spectra), so each component carries its own w. The net JES effect on the *summed* signal then automatically inherits the right κ_λ-dependence through the mixture. If one instead trained a single JES weight on a fixed-κ_λ total mixture (say the SM one), it would be silently wrong at $\kappa_\lambda = 5$ — a hidden λ-dependence smuggled in by implementation. That is why ATLAS trains $2 \times N_{\text{NP}} \times$*process* (signal components counted separately), and why the preliminary setup trains ratios per basis sample.

The **theory** NP is the one place where this clean structure genuinely fails *as prescribed* — because the uncertainty is only *given* at the total-signal level ($+4/-18\%$ *at the SM point*), with a κ_λ-dependent size and composition. Faithfulness then forces either the per-component decomposition (which restores α-only weights but needs the unpublished component-differential bands) or an explicitly κ_λ-dependent weight $w_\theta(x;\kappa_\lambda,\alpha_\theta)$ — that is exactly challenge C4.

1.4 "Pinned by two numbers, threaded through a fixed smooth family"

Question

What do you mean by "The entire α-dependence of each event's weight is then pinned by just those two numbers, threaded through a fixed smooth family"?

This sentence is about a **single event's weight as a function of** α — nothing to do with κ_λ. Fix one NP (say τES), one component s, one event x_e (Fig. 2):

1. **What is actually trained:** two classifiers — ($+1\sigma$ sample vs nominal) and (-1σ sample vs nominal). Via $s/(1-s)$ they yield two per-event numbers:

$$w_+(x_e) = \frac{p(x_e|\alpha{=}{+1})}{p(x_e|\alpha{=}0)}, \qquad w_-(x_e) = \frac{p(x_e|\alpha{=}{-1})}{p(x_e|\alpha{=}0)}. \quad (5)$$

   For the drawn event: $w_+ = 1.20$, $w_- = 0.85$ — "a $+1\sigma$ τES makes events *like this one* 20% more likely."

2. **What the fit needs:** $w(x_e,\alpha)$ at *every* α, because the minimizer explores α continuously (0.37, -1.6, $\ldots$) while profiling.

3. **The bridge:** through the three known points $(-1,\,w_-)$, $(0,\,1)$, $(+1,\,w_+)$, draw a curve from a **fixed, pre-chosen family** — the HistFactory "poly-exp": exponential in the tails (w_+^{α} for $\alpha > 1$, so $\alpha{=}{+2} \Rightarrow 1.20^2 = 1.44$; $w_-^{|\alpha|}$ for $\alpha < -1$), and for $|\alpha| < 1$ a 6th-order polynomial whose coefficients are *fully determined* by smoothly matching the tails (value, first and second derivative) at ± 1. **Zero free parameters remain.** At $\alpha = 0.5$ the curve gives $\approx w_+^{0.5} \approx 1.10$ for well-behaved anchors (the interior polynomial hugs the exponential).

That is the sentence: once $w_+(x_e)$ and $w_-(x_e)$ are known, the *entire function of* α for this event is decided — the event's kinematics enter only through where the two pins sit; the *shape of the path between and beyond the pins* is universal, identical for every event, every NP, every analysis.

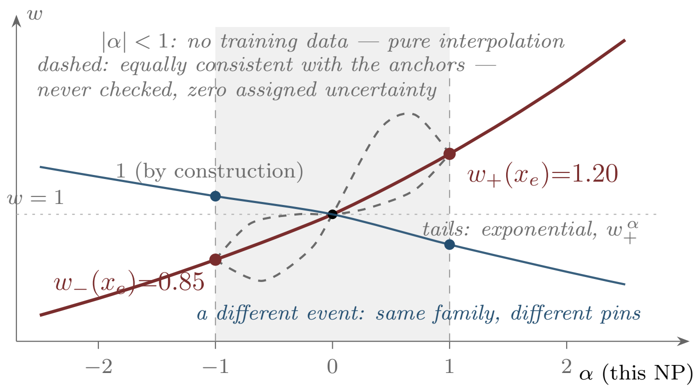

Figure 2: One event, one NP: two learned numbers ⇒ the entire α-curve. The x-dependence lives only in where the pins sit; the path between and beyond them is a universal, pre-chosen family. The dashed curves pass through the *same* three points — the training is equally happy with them — yet they would give different weights at $\alpha = 0.5$, different profiled likelihoods, and different κ_λ intervals.

The dashed alternatives are the point of the whole critique: the method does not just pick one curve, it picks one **and assigns zero uncertainty to the choice**. The R2 critique items of Sec. 3 now become concrete:

- **"Fixed order in α"** = this family. It is smooth and monotone between the anchors by construction — if the true per-event response is non-monotone (the dashed dipping curve; think of an event near a selection or SVfit threshold that migrates in and then out), the family *cannot represent it*, at any anchor values.
- **"Ignores kinematics for $|\alpha| < 1$"** = the shaded band. No simulation ever constrains the interior path. Sharper: if a sample were *generated* at $\alpha = 0.5$, this construction has **no slot to use it** — the family is already fully pinned by ± 1. (R4's joint parametric fit would simply add it to the loss; R3's GP would take it as another anchor *and* shrink its uncertainty band there.)
- **Per-NP independence** of these curves (one such plot per NP, multiplied together) is the factorization assumption; each pin-pair being trained in isolation is the "no joint likelihood ratios" point, with its C6 noise.

Connecting back to Sec. 1.3: κ_λ appears nowhere in Fig. 2 — that is the λ-independence made visible. The figure is drawn *per component*; the fit holds ∼dozens of such pin-pairs per event (one per NP × component), multiplies them, and mixes components with the known κ_λ coefficients. C4 is precisely the statement that for `THU_HH` no single such λ-blind curve can be faithful — the *pins themselves* would have to move as κ_λ changes, unless the theory band is first decomposed per component.

2 Profiling and the confidence interval

2.1 What "profiling" means

Question

Can you remind me what exactly 'profiling' means?

The full likelihood, with POI κ_λ and nuisances α (extended form, Gaussian constraints):

$$L(\kappa_\lambda, \alpha) \;=\; e^{-\nu_{\text{tot}}(\kappa_\lambda,\alpha)} \prod_e \nu(x_e; \kappa_\lambda, \alpha) \;\times\; \prod_m \mathcal{N}\big(a_m \,\big|\, \alpha_m,\, 1\big). \tag{6}$$

Profiling = re-fit the nuisances at each fixed κ_λ:

$$\hat{\hat{\alpha}}(\kappa_\lambda) \;=\; \arg\max_{\alpha} \; L(\kappa_\lambda, \alpha) \qquad \text{("double-hat": the systematics' \textit{best attempt to explain the data} given that } \kappa_\lambda\text{).} \tag{7}$$

The **profile likelihood ratio** and test statistic:

$$\lambda(\kappa_\lambda) \;=\; \frac{L\big(\kappa_\lambda,\; \hat{\hat{\alpha}}(\kappa_\lambda)\big)}{L\big(\hat{\kappa_\lambda},\; \hat{\alpha}\big)}, \qquad t(\kappa_\lambda) \;=\; -2\ln\lambda(\kappa_\lambda), \tag{8}$$

with the denominator the global best fit (single hats). Confidence intervals: $\mathrm{CI}_{68} = \{\kappa_\lambda : t \leq 1\}$, $\mathrm{CI}_{95} = \{\kappa_\lambda : t \leq 3.84\}$ (Wilks: $t \sim \chi^2_1$).

The intuition in one paragraph: **profiling is a structured "devil's advocate."** For every candidate κ_λ one asks: *if this were the true coupling, how well could the systematics conspire to explain my data anyway?* — and one lets them try (that is the re-fit, $\hat{\hat{\alpha}}(\kappa_\lambda)$). A systematic costs sensitivity **exactly to the extent it can mimic the POI** — the tilt of the likelihood contours in the (κ_λ, α) plane, which is the same $\rho(\kappa_\lambda, \alpha)$ as in the degeneracy toy. No tilt → profiling changes nothing; strong tilt (JES/τES on m_{HH}) → the parabola flattens and the CI widens. The width quoted is the *honest* one: it already includes every way the nuisances could have fooled you, weighted by their constraints.

2.2 Reading the widening off the picture: slice vs. shadow

Question

How to read off how the widening happens through profiling in the tilted case?

The key reframe: **the ellipse is a single fixed object** (the t=1 contour of the full 2-D likelihood). "Frozen" and "profiled" are two different ways of extracting a 1-D interval from it (Fig. 3):

- **Frozen** α = the **central horizontal slice**: walk left/right from the minimum along the line $\alpha = \hat{\alpha}$ and mark where you exit the contour.
- **Profiled** = the **shadow** (projection) of the whole ellipse onto the κ_λ axis: the extreme left/right points reached *anywhere* on the contour.

A slice can never be longer than the shadow — and they are equal exactly when the ellipse is axis-aligned. **The tilt is the entire effect.**

Now the mechanism, as a walk-through of the test point in the right panel. Pick a candidate κ_λ just outside the frozen interval (the open red circle):

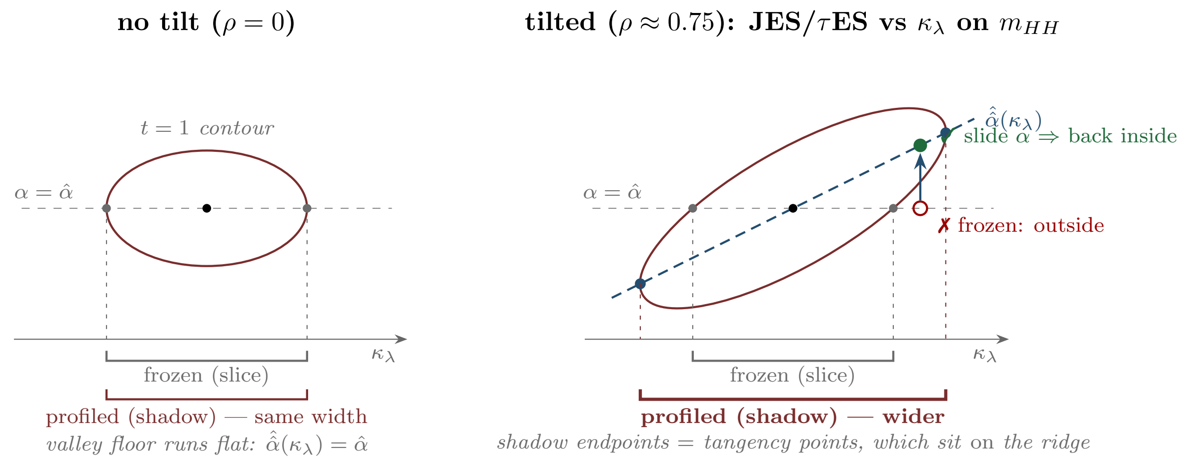

Figure 3: Frozen interval = central slice of the t=1 ellipse; profiled interval = its shadow (projection). They differ only when the ellipse is tilted.

1. **With α frozen**, the likelihood is evaluated at $(\kappa_\lambda^*, \hat{\alpha})$. That point sits *outside* the t=1 contour — this κ_λ appears excluded: the predicted m_{HH} shape sits in the wrong place.
2. **Profiling asks:** before excluding it, could a systematic explain the disagreement away? Slide *vertically* (κ_λ^* held fixed, α free) and descend the likelihood valley. Because the valley is **tilted** — a JES shift moves m_{HH} the same way a κ_λ shift does — descending really does recover likelihood: at $\hat{\hat{\alpha}}(\kappa_\lambda^*) \approx 0.6\sigma$ of JES, most of the mismatch is absorbed. The point lands *inside* the contour: **this κ_λ is no longer excludable**, because "slightly wrong JES" explains the data almost as well.
3. The interval must therefore extend until it reaches κ_λ values where *even the best α cannot rescue them* — and those are exactly the **tangency points**, where the vertical line just grazes the ellipse. That is why the shadow's endpoints sit on the ridge: the ridge is, by definition, the path of "best rescue," $\hat{\hat{\alpha}}(\kappa_\lambda)$.

The exact accounting (no approximation): at every κ_λ,

$$t_{\text{prof}}(\kappa_\lambda) \;=\; t_{\text{frozen}}(\kappa_\lambda) \;-\; t_{\text{recovered}}(\kappa_\lambda), \tag{9}$$

where $t_{\text{recovered}} \geq 0$ is the likelihood the nuisance wins back by sliding down. In the Gaussian case the recovered fraction is exactly ρ^2: the systematic can fake the ρ^2 "parallel component" of the κ_λ signature, and only the **orthogonal component, $(1 - \rho^2)$ of it, survives to constrain κ_λ:**

$$\boxed{t_{\text{prof}}(\kappa_\lambda) \;=\; (1 - \rho^2)\, t_{\text{frozen}}(\kappa_\lambda) \qquad \Longrightarrow \qquad \sigma_{\text{prof}} \;=\; \frac{\sigma_{\text{frozen}}}{\sqrt{1 - \rho^2}}} \tag{10}$$

The degeneracy-toy numbers *are* this formula

Exact for Gaussian likelihoods — and the project's degeneracy-toy results are this formula evaluated: with the 10-feature set $\rho = 0.65$, so $1/\sqrt{1 - 0.42} = 1.31\times$ wider; with m_{HH} alone $\rho \approx 0.997$ and the same expression gives $12.8\times$. When we said "dimensionality breaks the degeneracy," the mechanism was literally: extra features give κ_λ an effect component that JES *cannot* imitate, shrinking ρ, un-tilting the ellipse, and closing the gap between slice and shadow.

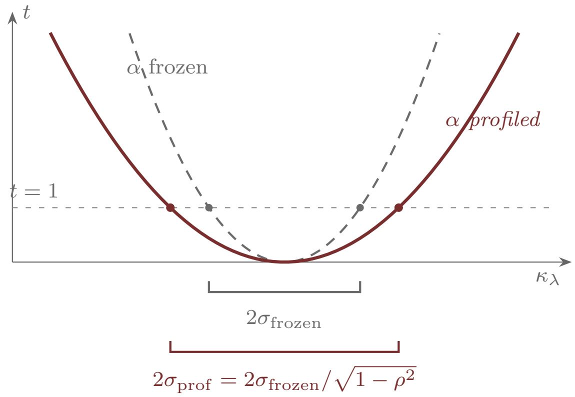

Figure 4: The profiled parabola is the frozen one flattened by exactly $(1-\rho^2)$; the $t \leq 1$ crossing moves out by $1/\sqrt{1-\rho^2}$ (drawn for $\rho = 0.75$, a factor 1.51).

2.3 The meaning of $\lambda(\kappa_\lambda) = L(\kappa_\lambda, \hat{\hat{\alpha}}(\kappa_\lambda)) \,/\, L(\hat{\kappa_\lambda}, \hat{\alpha})$

Question

Is there an intuition for the definition of $\lambda(\kappa_\lambda)$?

Read it as a **courtroom with the best possible defense**:

- **Numerator** — the hypothesis "the coupling is *this* κ_λ" gets a defense attorney who may tune every systematic in its favor (within the Gaussian-constraint penalties): *"the data look off because JES is* 0.6σ *high,* τ*ES* 0.3σ *low…"*. $L(\kappa_\lambda, \hat{\hat{\alpha}}(\kappa_\lambda))$ is the **best case that can possibly be made** for this κ_λ.

- **Denominator** — the best case for the **champion**: the globally best $(\hat{\kappa_\lambda}, \hat{\alpha})$, the hypothesis that explains the data best of all.

- λ is then a number in $[0, 1]$: *how close does this* κ_λ*'s best defense come to the champion's?* Near 1 → it explains the data essentially as well; there are no grounds to exclude it; it is in the interval. Small → even with every systematic maximally in its favor, it still fails; excluded. And $t = -2\ln\lambda$ converts "how much worse" onto the χ^2 scale (Wilks), which is what makes $t \leq 1 \leftrightarrow 68\%$ a universal calibration.

Why each ingredient is there:

- **Why a ratio at all?** An absolute likelihood value is meaningless — it depends on normalizations, units, and modeling constants that have nothing to do with κ_λ. Dividing by the global maximum cancels all of that and leaves a pure *relative* plausibility. It is also not arbitrary: for simple hypotheses the likelihood ratio is provably the most powerful test (Neyman–Pearson); λ is its canonical generalization to hypotheses with unknown nuisances (the generalized likelihood-ratio test).

- **Why maximize over α (double-hat) instead of freezing it?** Freezing α at the champion's $\hat{\alpha}$ rigs the trial — the nuisance values were chosen to fit the *champion*, so every other κ_λ gets judged with a defense tuned for someone else (that is the too-narrow slice interval of Fig. 3, and it under-covers). Profiling gives **each hypothesis its own best defense**, and that is precisely what buys the frequentist guarantee: the interval covers at the stated rate *no matter what the true* α *is* (asymptotically), because no κ_λ was excluded that some legitimate

α could have rescued.

- **Geometric reading:** the numerator, as a function of κ_λ, is the likelihood **along the ridge** — so $t(\kappa_\lambda)$ is simply *the height of the tilted valley's floor above the global minimum.* The profiled parabola of Fig. 4 is nothing but the valley floor seen edge-on from the κ_λ axis.

How the two questions join up

The $\hat{\hat{\alpha}}$ in the numerator of λ *is* the ridge in Fig. 3. Evaluating λ at some κ_λ *is* the act of sliding down to the ridge — the numerator does the rescuing, the denominator sets the standard, and the tilt (ρ) decides how much rescuing is possible, which is exactly how much the interval widens.

3 The four off-shell (R2) assumptions, and how they break per channel

Question

Can you elaborate on the meanings of these assumptions in the off-shell measurement (a fixed order in α; ignores kinematics for $|\alpha| < 1$; assumes factorization / no non-factorizable NPs; doesn't exploit joint likelihood ratios) — and how do they break for λ in the HH analysis, separately for $b\bar{b}\tau\tau$, $b\bar{b}\gamma\gamma$, $b\bar{b}b\bar{b}$?

Provenance. As Lesson 1 flags, that list is **the R4 group's critique of R2** — how the refinable/GOLLUM papers characterize the ATLAS approach. ATLAS would fairly reply that for clean 4ℓ these approximations were validated as adequate. What each means mechanically (all four are visible in Fig. 2):

1. **"Two classifiers per NP" + "fixed order in α."** R2 trains classifiers only at the $\pm1\sigma$ anchors: $w_+(x)$ and $w_-(x)$ per NP per process. The entire α-dependence of each event's weight is then **pinned by just those two numbers**, threaded through the fixed poly-exp family (Sec. 1.4). "Fixed order" = the *shape of the α-curve* is chosen a priori, not learned. If the true per-event response is non-monotone in α, or has genuine curvature the family cannot express, it is silently truncated — with **zero assigned uncertainty** for being wrong.

2. **"Ignores kinematics for $|\alpha| < 1$."** The x-dependence enters *only* through the two anchor weights. At $\alpha = 0.4$, event e's weight is a fixed interpolation of $w_+(x_e)$ and $w_-(x_e)$ — **no simulation at intermediate α ever informs it**. The hidden assumption: the *pattern* of the deformation across phase space at fractional α is the same as at $\pm1\sigma$, just scaled. When the response chain is nonlinear (reconstruction thresholds, migrations), different regions of phase space "turn on" at different α — and the interpolation cannot know.

3. **"Assumes factorization."** $w(x, \alpha) = \prod_m w_m(x, \alpha_m)$ — each NP acts independently on the density; no cross-terms. True when NPs touch different objects (lepton ES $\times$ lumi). False when two NPs feed the *same reconstructed quantity.*

4. **"Doesn't exploit joint likelihood ratios."** Every anchor classifier is trained **in isolation** — one (NP, direction, process) at a time. Nothing enforces consistency across them, there is no statistical pooling across variations, and each carries its *own* independent MC-stat noise (this is C6's multiplier). R4 instead fits **one** parametric surface $\log w(x, \alpha) = \sum_A \alpha_A \hat{\Delta}_A(x)$ to *all* variation samples jointly — shared statistics, consistent correlations, cross-terms included at chosen order.

Channel-by-channel breakage. ($b\bar{b}\tau\tau$ claims are grounded in this project's doc/CMS note; **$b\bar{b}\gamma\gamma$ and $b\bar{b}b\bar{b}$ are from the standard structure of those analyses — verify against the specific ATLAS/CMS analyses before quoting.**)

Assumption	$b\bar{b}\tau\tau$	$b\bar{b}\gamma\gamma$	$b\bar{b}b\bar{b}$
fixed order in α	**stressed** — τES propagates through MET→SVfit→ $m_{\tau\tau}$, a nonlinear likelihood-based estimator; JES drives category/selection migrations	mild — photon ES is sub-percent (linear regime); only the b-side JES has room to misbehave	**worst** — JES/JER can *flip the jet pairing* (which jets form h_1, h_2): per-event response is discontinuous in α, outside any smooth family
kinematics at $\lvert\alpha\rvert < 1$	**stressed** — τES is p_T- and decay-mode-dependent; the deformation *pattern* at fractional $\alpha \neq$ scaled anchor pattern	mild — small, nearly linear responses; anchors suffice	**stressed** — b-jet trigger turn-ons and pairing migrations cross thresholds at *different* α for different event subsets
factorization	**broken by construction** — JES×τES both feed $m_{HH} = m(b\bar{b}+\tau\tau)$ (C5); plus b-tag SFs are p_T-binned, so JES moves events across SF bins (JES×b-tag)	largely OK — PES×JES negligible (PES tiny); residual JES×b-tag mild	**worst** — 11 JES sources × JER × 4 b-tags all act on the *same four jets* and the same $m_{4b}/m_{h_1}/m_{h_2}$; pairing couples them jointly
joint LRs	**expensive to forgo** — ∼40 NPs × ∼7 processes × 2 anchors × ensemble ≈ thousands of independent nets, each noisy on the smallest $\pm 1\sigma$ samples (C6, C2)	fewer NPs; $\pm 1\sigma$ samples still finite — pooling helps, stakes lower	**expensive** — many NPs *plus* the background itself is a learned reweighting ($2b \to 4b$ / FvT-style NN with bootstrap ensembles); nobody has jointly treated a learned background + learned systematics
(channel wildcard)	data-driven QCD/fakes with no density (C3); `THU_HH` POI-coupling (C4)	continuum $\gamma\gamma$ fit in $m_{\gamma\gamma}$ sidebands → the "spurious-signal" NP is a **discrete model choice**, not a continuous α — does not map to morphing at all	dominant QCD estimate is itself a NN reweighting → its systematic is a nuisance *on a learned object*; C3 in its hardest form

Table 2: The four R2 assumptions across the three main HH channels.

The pattern worth internalizing

$b\bar{b}\gamma\gamma$ mostly inherits R2's assumptions safely (its hard problem is tiny yields, not morphing); **$b\bar{b}\tau\tau$ breaks factorization and stresses the fixed-form interpolation** (via the SVfit chain); **$b\bar{b}b\bar{b}$ breaks everything at once** (combinatorics make the response non-smooth, correlated, and the background is itself learned). That ordering — $\gamma\gamma$ easiest, $\tau\tau$ middle, $4b$ hardest — is also the order in which the R2 recipe ports without modification.

4 The upgrades: R3 (GP morphing) and R4 (refinable surrogate)

4.1 R3: Gaussian-Process morphing — mechanics, comparison, deployment for κ_λ

Question

Can you help me understand how R3 Gaussian-Process morphing works, and how it compares with what was done in off-shell, with our application to λ in mind? *(Context: R3 is led by the developer of the NSBI repo we've been working with.)*

Mechanics, single NP first. There are simulations at a few anchor settings of α — say $-2\sigma, -1\sigma, 0, +1\sigma, +2\sigma$. The quantity to morph (a bin's content ratio, or in the unbinned case a per-event weight / the coefficient functions of the weight) is an unknown function $\Delta(\alpha)$. Instead of *assuming* poly-exp, place a **Gaussian-Process prior** on it — "Δ is some smooth function, with correlations over α set by a kernel $k(\alpha, \alpha')$" — and **condition on the anchors**. The posterior gives two things at every α:

- a **mean** — the interpolated morphing the fit uses (playing exactly the role poly-exp played), and
- a **covariance** — *how uncertain the interpolation itself is*: small at anchors, growing between and especially **beyond** them.

That covariance is the unique feature: it enters the fit as extra constrained parameters (γ) — so if the fit pulls a nuisance into a region where the morphing is poorly determined, the likelihood **automatically pays a penalty** instead of trusting an assumption. This is "honest morphing" — study #9 of the challenges note, and the direct mitigation of the *tighter-but-mis-covering* failure mode.

Multi-NP: two R2 assumptions die at once. The GP's input can be the *vector* α, and the anchors **need not sit on the axes** — a simulated sample with JES and τES shifted *together* is just another anchor point. The kernel then learns the joint response surface directly: **no factorization assumed** (C5 handled by data, not by ansatz) and **no fixed order** (non-parametric). A subtle bonus relevant to C6: anchor points are themselves noisy (finite $\pm 1\sigma$ MC), and GP regression naturally takes per-anchor noise — so the MC-stat of the *variation samples* feeds coherently into the interpolation band rather than being handled by a separate bolted-on estimator.

For κ_λ specifically. Challenge C1 says morphing accuracy *directly sets* the κ_λ uncertainty, because the systematics deform the observable that carries the signal. That is precisely the regime where an *assumed-perfect* interpolation is dangerous and an honest band is worth its width. The pragmatic deployment is not "GP everything" — it is: keep the R2 skeleton, and **swap poly-exp $\rightarrow$ GP for the top-impact, cross-term-prone NPs** (the JES sources and τES components feeding m_{HH}), requesting a handful of off-axis anchor samples for exactly those pairs. Note the GP does *not* by itself solve C4 — the POI-coupling of `THU_HH` needs the per-component decomposition first; GP can then morph each component honestly.

The sociology is a real planning input. R3's Sandesara is both the off-shell lead author and the developer of the upstream toolkit this project builds on, with `pyhf-gpsys` headed for the standard stacks. So **R3 is the low-friction, in-ecosystem upgrade path** (likely with direct

	R2 off-shell (poly-exp)	R3 (GP)
functional form in α	fixed family, pinned by $\pm 1\sigma$	learned from anchors, non-parametric
cross-terms (JES$\times\tau$ES)	excluded (factorized product)	included via off-axis anchors + joint kernel
interpolation error	**assumed zero**	**quantified** (posterior cov $\to$ constrained γ)
anchor MC-stat	separate estimator (the contested C6)	enters as GP observation noise
cost / risk	cheap, rigid	kernel choice = injected bias; needs more anchor samples (esp. off-axis); pre-publication

Table 3: Off-shell interpolation vs GP morphing, term by term.

support in the very framework in use), while R4 is an external stack — which sets up the next question.

4.2 R4 is ***not*** generative (not normalizing flows) — and why that is good news

Question

R4 is a generative-based approach (normalizing flows) — does that apply to ours (training a classifier to learn the density ratio) at all? *(Context: R4 is by another group, despite seeming to have many advantages.)*

This premise deserves a careful correction, because it changes the applicability answer completely. Checked against the lessons *and* the sources: R4 — "Refinable modeling for unbinned SMEFT analyses" (MLST 6 015007) and its application paper "Unbinned inclusive cross-section measurements with machine-learned systematic uncertainties" (PRD 112 052006) — is **not** flow-based or generative. It learns **machine-learned surrogates of the likelihood *ratio***, with the coefficient functions of the ansatz

$$\log \frac{\sigma(\alpha)}{\sigma(0)}\Big|_x = \sum_A \alpha_A \, \hat{\Delta}_A(x) \qquad (A \text{ ranging over linear, quadratic, and cross-terms}) \tag{11}$$

fit by a **tree-boosting algorithm** (and NN variants) trained with cross-entropy-style losses on the variation samples. That is *discriminative ratio estimation* — the same statistical trick as the CARL classifiers used throughout this project — just organized as one **joint, α-parametric** fit instead of independent anchor pairs.

So the answer to "does it apply to ours at all?" is: **it applies maximally — it *is* our approach, upgraded.** Porting R4's idea into the pipeline means replacing

$$\{2 \text{ independent classifiers per NP}\} + \{\text{fixed poly-exp bridge}\}$$

with

$$\{\text{one set of coefficient networks } \hat{\Delta}_A(x) \text{ per process, fit jointly to } \textit{all} \text{ variation samples, with cross-terms } (A = \text{pairs}) \text{ included at whatever order is affordable — "refinable"}\}.$$

No generative modeling, no flows, no change to the reference-hypothesis trick or the fit — just a better-parameterized ratio head trained with machinery already in place.

Where the flows association likely came from (worth untangling, since both live in this program): **flows appear in the *background-density* thread, not the systematics thread** — ABCDnn (the neural autoregressive flow learning a data-driven QCD/fakes density, challenge C3) and the FlowNF-style toy cited once alongside GOLLUM in the C4 verdict box of the challenges note. Those are generative because their job is to *manufacture a density that does not exist.* R4's job is to *deform a density you already have* — a ratio problem, solved discriminatively.

Resolving R3-vs-R4 without choosing

R4 is an idea implementable inside our own stack (a joint parametric ratio head — no dependency on GOLLUM), while **R3 is an ecosystem component** likely to arrive with native support in the framework in use. The hybrid already on record — R2 skeleton + R4-style joint/cross-term ansatz where factorization breaks + R3-style GP band where interpolation honesty matters most — uses each group's contribution where it is strongest, without adopting either wholesale.

5 Synthesis: mapping this Q&A onto the challenge list

One box to remember

- **C1 (κ_λ–systematic degeneracy)** = the *tilt* ρ of Sec. 2.2; the widening is exactly $1/\sqrt{1-\rho^2}$; dimensionality shrinks ρ.
- **C2 (NP scaling)** = the count of Family-2 objects (Table 1): $2 \times N_{\mathrm{NP}} \times N_{\mathrm{comp}} \times$ ensemble.
- **C4 (POI-coupled theory)** = the green/red boundary of Sec. 1.1 leaking; in the pin picture, the pins themselves would have to move with κ_λ.
- **C5 (cross-terms)** = multiplying per-NP pin-curves (Fig. 2); relaxed by R4's joint ansatz or R3's off-axis anchors.
- **C6 (MC-stat of learned ratios)** = each pin-pair trained in isolation carrying its own noise; relaxed by joint training (R4), GP anchor-noise (R3), or an analytic estimator (wifi).

References

[1] ATLAS Collaboration (J. Sandesara *et al.*), *An implementation of neural simulation-based inference for parameter estimation in ATLAS*, arXiv:2412.01600, Rep. Prog. Phys. 88 (2025). The R2 recipe: per-event factorized weights, $\pm 1\sigma$ anchor classifiers, poly-exp interpolation.

[2] R. Schöfbeck *et al.*, *Refinable modeling for unbinned SMEFT analyses*, arXiv:2406.19076, MLST 6 015007 (2025). The R4 method: tree-boosted, α-parametric likelihood-ratio surrogates (discriminative, *not* flows).

[3] R. Schöfbeck *et al.*, *Unbinned inclusive cross-section measurements with machine-learned systematic uncertainties*, arXiv:2505.05544, PRD 112 052006 (2025). R4 tested (GOLLUM), incl. on the FAIR Universe benchmark.

[4] K. Cranmer, M. Feickert, L. Heinrich, A. Held, J. Sandesara, *HistFactory v2 / Gaussian-Process systematic morphing*, SBI Blueprint (Feb 2026); https://github.com/pyhf/pyhf-gpsys. The R3 mechanism: GP over anchors, posterior covariance as interpolation uncertainty.

[5] *New challenges for an NSBI measurement of κ_λ in $HH \to b\bar{b}\tau\tau$*, this workspace, `2026-06-20-klambda-nsbi-vs-offshell-challenges.pdf` — challenge labels C1–C6, the maturity/risk matrix, and the width and compute analyses referenced here.